Alexandra Neagu

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Education

Imperial College London *Sep 2024 – Sep 2028*

PhD Applied AI

- Advisor: Dr Peter B. Johnson, Dr Rhodri Nelson
- Researching: LLM-chatbot alignment with educational values, Human-computer interaction, AI-assisted STEM self-study

Imperial College London *Oct 2020 – Jun 2024*

MEng Electrical and Information Engineering, First Class Honours

- Imperial College President's Undergraduate Scholarship for academic excellence
- Final Year Project: designed and ran a 25-participant user study on LLM-generated, skill-tailored hints within a custom Unity puzzle game: AI assistance improved task efficiency but did not improve learning gains

Industry Experience

Lambda Feedback *Sep 2024 – present*

Forward Deployed Scientist: LLM-Assisted Self-Study (PhD research) *London, UK*

- Design, build and evaluate LLM-powered self-study tools on Lambda Feedback (2,000+ active students yearly, 70+ STEM modules); shaped behaviour of LLM chatbots based on user research and analysis of real interaction data, reaching 200+ students since launch in Spring 2025
- Co-designed μEd API, a platform-independent API for educational tools to share functionality, now being adopted by Imperial, NTU Singapore, TU Munich and ETH Zürich

Amazon Web Services *Apr – Sep 2023*

Professional Services Consultant Intern *London, UK*

- Presented and defended technical decisions in architecture reviews with senior consultants across EMEA and the US
- *Global monitoring & alerting for AWS's virtual-desktop service:* built back-end automation, dashboards and alarms across the global EC2 fleet; agreed metrics and thresholds with service owners, letting the team right-size the fleet and flag unused or suspicious instances
- *Text-to-Speech on AWS Polly (Emirates iXR):* designed an end-to-end serverless pipeline generating voice audio for 3D avatars in employee training, shipped as a reusable Terraform deployment with a configurable React/TypeScript UI

Amazon Web Services *Jul – Sep 2022*

Professional Services Consultant Intern *London, UK*

- Turned AWS's Landing Zone Assessment into a customer- and partner-facing offering adopted as standard practice, cutting consultant assessment time by ∼20% (∼80h workflow) via automated report generation

Skills

Technical: LLM app design & evaluation, agentic workflows; Python, TypeScript/React, SQL, PyTorch; AWS (ML Speciality & Solutions Architect certified), GCP, Docker, Pulumi

Languages: English & Romanian (native), German (B1-B2), Mandarin (HSK 4-5)

Publications

Rethinking Scaffolding in LLM Tutors: The Interactional Mismatch Between Benchmarks and Real-World Deployments. **Neagu, A.** et al. *ICML Pluralistic Alignment Workshop 2026*

'How Do I. . . ?': Procedural Questions Predominate Student-LLM Chatbot Conversations. **Neagu, A.** et al. *arXiv 2026*

μEd API: Towards a Shared API for Education Microservices. Sölch, M., **Neagu, A.** et al. *ACM L@S 2026*

Chatbots for Dialogic Feedback during Self-Study: The Importance of Contextual Information. **Neagu, A.** et al. *EERN 2025*

Workshops & Tutorials

Co-organiser **PEAF 2026: Pedagogical Evaluation of Automated Feedback** — full-day workshop *Festival of Learning 2026*

Co-organiser **Build and Deploy Microservices for Automated Feedback** — half-day tutorial *Festival of Learning 2026*

Other Leadership Experience & Interests

President, Dance Club – Imperial College London *2026 – 2027*

- Lead a committee of 10+ to run weekly classes, workshops, and socials for 200+ active members

Vice President Industry, EESoc – Imperial College London *2021 – 2023*

- Managed a team of 4 to build industry partnerships and negotiate sponsorships; led one of Imperial's largest student-led Careers Fairs and Intro-to-Industry events

Other Activities *ongoing*

- Participated in research summer schools: LearnLab (Carnegie Mellon University, 2025), SEFI (Aalborg University, 2026)
- Participated or volunteered at Hackathons every year since 2021, building projects across various domains and technologies (ICHack 2021-2026, AWS Interns Hack 2023, WSET Hack-her-thon 2021-2022)